

Home SOC Lab

Security Operations Center Guide

By : FAIZAN MOHAMMED

github.com/fai2an/HomeSOCLab

```
Kali Linux (Attacker + Agent) --SSH Brute Force--> Ubuntu Server (Wazuh Manager)
|
Windows Browser (Wazuh Dashboard)
```

Tool	Version	Role
VMware Workstation	Latest	Virtualization platform
Kali Linux	2025.3	Attacker and Wazuh Agent
Ubuntu Server	25.04	Wazuh Manager host
Wazuh SIEM	4.7.5	Security monitoring
ncrack	0.7	Brute force simulation
OpenSSH	Latest	Attack target service

Table of Contents

Section	Title
1	Introduction and Lab Overview
2	Prerequisites
3	VM Configuration
4	Network Setup
5	Wazuh SIEM Installation
6	Wazuh Agent Setup on Kali
7	Attack Simulation
8	Dashboard Analysis
9	Alerts and Rules Explained
10	MITRE ATT&CK Mapping
11	Incident Report
12	Skills and Lessons Learned

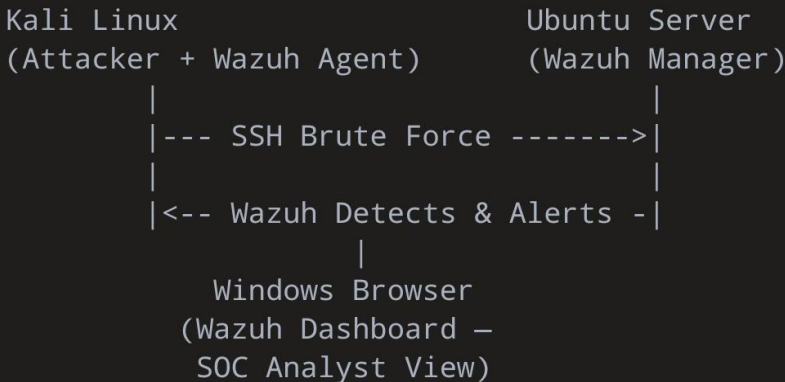
Section 1: Introduction and Lab Overview

1.1 What is this Project?

This is a home Security Operations Center lab built from scratch on a single laptop. The goal was to understand how real SOC analysts detect and respond to attacks, not just read about it in theory.

A real SIEM (Wazuh) was deployed, a monitored endpoint was connected, an actual brute force attack was launched against the lab, watched it get detected in real time and documented everything as a professional incident report.

1.2 Lab Architecture



1.3 Why I Built This

Every SOC Analyst job description mentions SIEM experience. Instead of just watching tutorials, building it from scratch was the better approach. This project provided more learning in one weekend than months of passive study, especially around how alerts are generated, what MITRE ATT&CK; looks like in practice and how to think like a defender.

1.4 Target Roles

Role	Relevance to This Project
SOC Analyst	SIEM deployment, alert analysis, incident reporting
Security Operations	Real-time threat detection and response
Blue Team	Defensive security, monitoring, log analysis

Section 2: Prerequisites

2.1 Hardware Requirements

Component	Minimum	Recommended
RAM	8 GB	16 GB
Storage	60 GB free	100 GB free
CPU	4 cores	8 cores
OS	Windows 10/11	Windows 10/11

2.2 Software Required

Software	Version	Purpose
VMware Workstation	Latest	Run virtual machines on Windows host
Kali Linux	2025.3 x64	Attacker machine and Wazuh Agent
Ubuntu Server	25.04 LTS	Wazuh Manager and SIEM host
Wazuh SIEM	4.7.5	Security monitoring platform
ncrack	0.7	SSH brute force simulation tool

2.3 Download Links

Software	URL	Reference
VMware Workstation	vmware.com/products/workstation-pro	Install guide (free):youtu.be/XzD8JIAOk2I
Kali Linux	kali.org/get-kali	Download the VMware image
Ubuntu Server	ubuntu.com/download/server	Download 25.04 LTS ISO

2.4 Knowledge Required

Basic Linux command line (navigating directories, running commands)

Basic understanding of networking (IP addresses, ports)

No prior SIEM experience needed

Section 3: VM Configuration

3.1 Overview

Two virtual machines are required for this lab. Each machine plays a specific role:

VM	Operating System	Role
VM 1	Kali Linux 2025.3	Attacker and Wazuh Agent
VM 2	Ubuntu Server 25.04	Wazuh Manager (SIEM)

3.2 Kali Linux VM Settings

Open VMware Workstation, right-click your Kali VM and select Settings. Configure as follows:

Setting	Value
Memory	2048 MB (2 GB)
Processors	1 processor, 2 cores per processor
Hard Disk	80 GB (existing, no change needed)
Network Adapter	NAT (for installation phase)

3.3 Ubuntu Server VM Setup

Create a new VM in VMware: File -> New Virtual Machine -> Typical -> Select Ubuntu ISO. Configure as follows:

Setting	Value
Memory	2048 MB (2 GB)
Processors	1 processor, 2 cores per processor
Hard Disk	20 GB
Network Adapter	NAT (for installation phase)
VM Name	Ubuntu-SOC-Target

3.4 Ubuntu Server Installation Settings

During the Ubuntu installation process, select the following options:

Installer Screen	What to Select
Language	English
Keyboard Layout	English (US)
Type of Install	Ubuntu Server (NOT minimized)
Network	Leave default (DHCP)
Storage	Use entire disk, no LVM
Username	analyst
Server Name	soc-target
Password	Your choice (remember it)
OpenSSH Server	YES (check this box)
Featured Snaps	Skip, select nothing

Note: Always check the OpenSSH Server option during installation. This is critical for the attack simulation later.

3.5 Network Adapter Plan

During setup both VMs use NAT to download packages from the internet. Once all installations are complete, both VMs remain on NAT as they share the same subnet automatically.

Phase	Network Setting	Reason
Installation and setup	NAT	Needs internet to download packages
Attack simulation	NAT	Both VMs on same subnet automatically

Section 4: Network Setup

4.1 Overview

Before installing Wazuh, both VMs must be able to communicate with each other. This section verifies network connectivity between Kali and Ubuntu.

4.2 Get IP Addresses

Boot both VMs and run the following command on each:

```
# Run on both Ubuntu and Kali

ip a

# Look for the line that says inet under ens33 or eth0

# That is your VM IP address
```

VM	Interface	Example IP Address
Ubuntu Server	ens33	192.168.44.129
Kali Linux	eth0	192.168.44.128

Note: Your IP addresses may be slightly different but both should start with the same subnet such as 192.168.x.x

4.3 Ping Test

From the Kali terminal, ping Ubuntu to confirm both VMs can communicate:

```
# Run on Kali terminal

ping -c 4 192.168.44.129

# Expected output:
# 64 bytes from 192.168.44.129: icmp_seq=1 ttl=64 time=0.45 ms
# 64 bytes from 192.168.44.129: icmp_seq=2 ttl=64 time=0.38 ms
# 64 bytes from 192.168.44.129: icmp_seq=3 ttl=64 time=0.41 ms
# 64 bytes from 192.168.44.129: icmp_seq=4 ttl=64 time=0.39 ms
#
# 4 packets transmitted, 4 received, 0% packet loss
```

0% packet loss confirms both VMs are on the same subnet and can communicate successfully.

4.4 Lab Network Summary

Machine	IP Address	Role	Network
Ubuntu Server	192.168.44.129	Wazuh Manager	NAT
Kali Linux	192.168.44.128	Attacker and Agent	NAT
Windows Host	Your PC IP	SOC Analyst Dashboard	Host

4.5 Troubleshooting

Issue	Fix
Different subnets (192.168.44.x vs 192.168.56.x)	Make sure both VMs are set to the same network type in VMware
Request timed out	Check if both VMs are powered on
100% packet loss	Restart network adapter on both VMs

Section 5: Wazuh SIEM Installation

5.1 What is Wazuh?

Wazuh is a free, open-source SIEM (Security Information and Event Management) platform used by real SOC teams worldwide. It collects logs from machines, detects threats and raises alerts in real time. Installing it from scratch demonstrates real hands-on security experience.

5.2 Update Ubuntu

Before installing Wazuh, update Ubuntu packages. Make sure the Network Adapter is set to NAT for internet access:

```
# Run on Ubuntu

sudo apt-get update && sudo apt-get upgrade -y

# Install curl (needed to download Wazuh)

sudo apt install curl -y
```

5.3 Install Wazuh (All-in-One)

This single command installs Wazuh Manager, Indexer and Dashboard automatically:

```
# Download and run Wazuh installer

curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh

sudo bash wazuh-install.sh -a --ignore-check

# --ignore-check flag is needed for Ubuntu 24.04 and 25.04

# Installation takes approximately 10 to 15 minutes
```

Note: At the end of installation you will see admin credentials. Save the password immediately. You cannot recover it easily.

5.4 Reset Admin Password

If you lose the password or need to reset it, run:

```
sudo /usr/share/wazuh-indexer/plugins/opensearch-security/tools/
wazuh-passwords-tool.sh -u admin -p YourNewPassword

# Password must contain: uppercase, lowercase, number and symbol

# Example: SOClab+Admin123
```

5.5 Restart Wazuh Services

If the dashboard is not loading or API shows an error:

```
sudo systemctl restart wazuh-manager
sudo systemctl restart wazuh-indexer
sudo systemctl restart wazuh-dashboard

# Wait 60 seconds then refresh the browser

# Verify all services are running:
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
```

5.6 Access Wazuh Dashboard

Open a browser on your Windows host machine and navigate to:

```
# Open browser on Windows host
https://192.168.44.129

# Login credentials:
Username: admin
Password: (your set password)

# Browser will show a security warning
# Click Advanced then Proceed to continue
# This is normal for self-signed certificates
```

Section 6: Wazuh Agent Setup on Kali

6.1 What is the Wazuh Agent?

The Wazuh Agent is a lightweight program installed on a monitored machine. It collects logs, monitors file changes and detects suspicious activity, then sends everything to the Wazuh Manager in real time.

Component	Location	Function
Wazuh Manager	Ubuntu Server	Receives logs, analyzes them, generates alerts
Wazuh Agent	Kali Linux	Collects and sends logs to the Manager
Wazuh Dashboard	Windows Browser	Displays alerts for SOC analyst review

6.2 Dual Role of Kali Linux

In this lab Kali Linux plays two roles simultaneously:

Role	Tool Used	What it Does
Attacker	ncrack	Launches SSH brute force attack against Ubuntu
Agent	Wazuh Agent	Reports its own activity and logs to Wazuh Manager

Info: In a real SOC environment these would be separate machines. The attacker would be an external threat while the agent would only be installed on company endpoints. Both roles are combined in Kali to keep the lab running on a single laptop.

6.3 Install Wazuh Agent on Kali

Run this command on Kali Linux. Make sure Kali is on NAT for internet access:

```
# Download and install Wazuh Agent on Kali
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/
wazuh-agent_4.7.5-1_amd64.deb

sudo WAZUH_MANAGER='192.168.44.129' \
WAZUH_AGENT_NAME='kali-attacker' \
dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb
```

6.4 Fix Manager IP in Config

The installer may use a placeholder IP. Fix it with:

```
# Fix Manager IP automatically

sudo sed -i 's/MANAGER_IP/192.168.44.129/g' /var/ossec/etc/ossec.conf

# Verify the change was applied:

sudo grep -i address /var/ossec/etc/ossec.conf

# Should show: 192.168.44.129
```

6.5 Start Wazuh Agent

```
sudo systemctl daemon-reload

sudo systemctl enable wazuh-agent

sudo systemctl start wazuh-agent

# Verify agent is running:

sudo systemctl status wazuh-agent

# Look for: Active: active (running)
```

6.6 Verify Agent on Dashboard

On Windows browser, open the Wazuh Dashboard and check:

```
1. Open https://192.168.44.129
2. Login with admin credentials
3. Check the main dashboard overview:

Total agents: 1

Active agents: 1

Disconnected agents: 0
```

Active agents: 1 confirms Kali is successfully connected to Wazuh Manager on Ubuntu.

Section 7: Attack Simulation

7.1 What is a Brute Force Attack?

A brute force attack is when an attacker tries many username and password combinations to gain access to a system. It is one of the most common attacks SOC analysts see in real environments. In this lab ncrack is used to simulate this attack against Ubuntu SSH service.

7.2 Verify SSH on Ubuntu

```
# Run on Ubuntu to check SSH status

systemctl status ssh

# If not running, install and start it:

apt-get install openssh-server -y

systemctl start ssh

systemctl enable ssh
```

7.3 Test SSH from Kali

```
# Run on Kali to test SSH connectivity

ssh analyst@192.168.44.129

# Type yes when asked about fingerprint

# Type any wrong password

# You should see: Permission denied

# This confirms SSH is reachable from Kali
```

7.4 Create Password Wordlist

Create a file containing passwords for ncrack to try:

```
# Run on Kali to create the wordlist
nano ~/passwords.txt

# Type these passwords one per line:
# 123456
# password
# admin
# root
# toor
# kali
# test
# letmein
# qwerty
# abc123
# analyst
# SOClab123
# welcome
# password123

# Save: Ctrl+X then Y then Enter

# Verify the file was created:
cat ~/passwords.txt
```

7.5 Launch Brute Force Attack

```
# Run on Kali to launch SSH brute force
ncrack -u analyst -P /home/kali/passwords.txt -p 22 192.168.44.129 -v

# Expected output:
# Starting Ncrack 0.7 ( http://ncrack.org )
# ssh://192.168.44.129:22 finished.
# Ncrack done: 1 service scanned in 18.02 seconds
# Probes sent: 9
```

Note: Open the Wazuh Dashboard on Windows browser simultaneously. Alerts will appear within seconds of the attack starting.

Section 8: Dashboard Analysis

8.1 Navigating to Security Events

1. Open `https://192.168.44.129` in browser
 2. Login with admin credentials
 3. Click hamburger menu (three lines) top left
 4. Click Wazuh
 5. Click Modules
 6. Click Security Events
- # For MITRE ATT&CK; view:
1. Hamburger menu -> Wazuh -> Modules
 2. Click MITRE ATT&CK;

8.2 Understanding the Graphs

Graph	What It Shows	What to Look For
Alerts Evolution Over Time	Alert count per 30-minute interval over last 24 hours	Spike at the exact time of attack. Flat line means no activity
Alert Level Evolution	Severity levels over time ranging from Level 3 to Level 15	Purple: Level 7 High Green: Level 8 Critical Dark: Level 10 Very Critical
Attacks by Technique	MITRE ATT&CK techniques detected and their frequency	Password Guessing bar SSH bar Brute Force bar
Top Tactics by Agent	Which machine triggered which attack tactic	Kali bar vs Ubuntu bar Credential Access is your attack
MITRE Techniques by Agent	Donut chart showing techniques per monitored endpoint	Kali slice vs Ubuntu slice Size shows frequency of detection

8.3 Alert Severity Levels

Level	Severity	Meaning
1 to 3	Low	Informational: normal activity, login sessions opened
4 to 5	Medium	Suspicious: failed logins, service changes
6 to 7	High	Serious: SSH attempts, anomaly detected by rootcheck
8 to 9	Critical	Auth attempts exceeded, new users added to system
10+	Very Critical	Brute force confirmed, multiple rapid failures detected

Section 9: Alerts and Rules Explained

9.1 Attack-Related Rules

Rule ID	Description	Level	Count	Meaning
5712	sshd: brute force trying to get access	10	1	Wazuh confirmed brute force attack
5551	PAM: Multiple failed logins in short time	10	1	Rapid fire login attempts detected
5758	Maximum authentication attempts exceeded	8	1	SSH blocked further login attempts
5710	SSH login: non-existent user attempt	5	25	Wrong usernames were tried by attacker
5503	PAM: User login failed	5	9	Matches exactly 9 ncrack probes sent
2502	syslog: Password missed multiple times	10	1	Multiple wrong passwords recorded

9.2 System Activity Rules

Rule ID	Description	Level	Count	Meaning
2904	Dpkg package half configured	7	49	Wazuh Agent package installation activity
510	Host-based anomaly detection	7	48	Wazuh rootcheck scanner running normally
2902	New dpkg package installed	7	46	curl and wazuh-agent packages installed
5501	PAM: Login session opened	3	21	Each time Ubuntu terminal was opened
5402	Successful sudo to ROOT executed	3	13	Every sudo command you ran was logged
501	New Wazuh agent connected	3	1	Kali agent successfully connected to Manager
502	Wazuh server started	3	4	Manager was restarted 4 times during setup
521	Possible kernel level rootkit	11	1	FALSE POSITIVE: Kali hacking tools look suspicious to scanner

Info: Rule 521 (Possible Rootkit Level 11) is a false positive. Kali Linux contains many security testing tools that trigger Wazuh rootkit scanner. This is expected in a security testing lab.

Section 10: MITRE ATT&CK; Mapping

10.1 What is MITRE ATT&CK;?

MITRE ATT&CK; is a globally recognized framework that categorizes real-world attack techniques used by threat actors. Wazuh automatically maps detected alerts to this framework, helping analysts understand the attacker goal and technique without reading every individual log.

10.2 Attack Mapping

MITRE Component	Value	Description
Tactic	Credential Access	The attacker goal was to steal login credentials
Technique	Brute Force	The method used was trying many passwords
Sub-technique	T1110.001: Password Guessing	Specific type: guessing known common passwords
Technique ID	T1110	Official MITRE ATT&CK identifier for brute force
Detection Rules	5712, 5551, 5710	Wazuh rules that caught the attack

10.3 All Tactics Detected

Tactic	Meaning	Detected?
Credential Access	Stealing passwords	YES: Main attack tactic
Defense Evasion	Avoiding detection	YES: Kali tools triggered this
Privilege Escalation	Gaining higher access	YES: Sudo activity logged
Lateral Movement	Moving between systems	YES: SSH connection attempts
Persistence	Maintaining access	YES: Agent startup activities
Initial Access	First entry into the system	YES: SSH brute force attempt

Section 11: Incident Report

A professional incident report documents what happened, when it happened, how it was detected and what should be done to prevent it. This is a core SOC Analyst skill required in real security operations.

SOC INCIDENT REPORT

Incident ID	IR-2026-001	Severity	High (Level 10)
Date	April 26, 2026	Status	Detected and Contained
Attack Type	SSH Brute Force	Tool Used	ncrack v0.7

Section	Details
1. Summary	A brute force SSH attack was detected against Ubuntu Server (192.168.44.129) originating from Kali Linux (192.168.44.128). The attack involved 9 SSH login probes using a password wordlist. No successful login was achieved.
2. Affected Systems	Target: Ubuntu Server IP: 192.168.44.129 Service: SSH Port 22 Attacker: Kali Linux IP: 192.168.44.128 Tool: ncrack v0.7
3. Attack Timeline (GST)	00:14 - ncrack launched from Kali Linux 00:14 - 9 SSH login probes sent to Ubuntu Server 00:14 - Wazuh Rule 5710 fired (25 alerts generated) 00:15 - Rule 5712 Level 10 confirmed brute force attack 00:15 - Rule 5551 fired: multiple failures in short time 00:15 - Attack ended: ncrack exhausted the wordlist
4. Detection	SIEM: Wazuh v4.7.5 Primary Rule: 5712 SSH Brute Force MITRE ATT&CK:: T1110 Brute Force Tactic: Credential Access
5. Evidence	Wazuh MITRE ATT&CK; dashboard screenshots Official Wazuh Security Events Report PDF ncrack terminal output showing 9 probes sent Alert evolution graph showing spike at attack time
6. Response Actions	Attack detected automatically by Wazuh SIEM Source IP identified: 192.168.44.128 (Kali Linux) Attack contained within isolated lab network No successful authentication was achieved
7. Recommendations	1. Install fail2ban to auto-block IPs after 3 failed attempts 2. Disable password authentication and use SSH keys only 3. Change SSH port from default 22 to a custom port 4. Implement IP whitelisting for SSH access 5. Set strong password policy: 14 characters minimum with complexity 6. Enable real-time email alerts for Level 8 and above events

8. Conclusion

The brute force attack was successfully detected by Wazuh SIEM within seconds of initiation. The attacker attempted 9 password combinations against the SSH service. No successful login was achieved. All recommended hardening measures should be implemented immediately to reduce the attack surface.

Section 12: Skills and Lessons Learned

12.1 Skills Demonstrated

Skill	Evidence	Relevant Role
Virtualization	VMware: two VMs configured	SOC / SysAdmin
Linux Administration	Ubuntu and Kali configuration	All roles
SIEM Deployment	Wazuh installed from scratch	SOC Analyst
Endpoint Monitoring	Wazuh Agent connected to Manager	SOC Analyst
Attack Simulation	ncrack brute force launched	VAPT
Threat Detection	Level 10 alerts generated	SOC Analyst
Log Analysis	Rules 5712 and 5551 analyzed	SOC Analyst
MITRE ATT&CK	T1110 Brute Force mapped	SOC Analyst
Incident Reporting	Professional report written	SOC Analyst
Timezone Analysis	GST vs EDT difference identified	SOC Analyst
False Positive ID	Rule 521 correctly identified	SOC Analyst
Network Configuration	NAT and Host-only switching	Network Engineer
Troubleshooting	Multiple real issues resolved	All roles

12.2 Key Lessons Learned

A SIEM generates alerts based on rules. Understanding those rules is what separates a good analyst from someone who just watches dashboards.

MITRE ATT&CK; is not just a theoretical framework. Seeing T1110 light up in real time after an attack makes it click in a way reading never does.

Timezone differences matter in incident response. Logs recorded in EDT look completely different from local GST time.

False positives are a real challenge in SOC work. Rule 521 (rootkit alert) firing because of Kali tools is a perfect example of why analysts need context.

Troubleshooting is a core skill. The Wazuh installation, agent configuration and network issues all required real problem solving.

Building something from scratch is the fastest way to learn. Every error message and failed command taught more than any tutorial.

12.3 Completion Status

Phase	Task	Status
Phase 1	VMware Lab Setup	Complete
Phase 1	Kali Linux Configuration	Complete
Phase 1	Ubuntu Server Installation	Complete
Phase 1	Network Connectivity Verified	Complete
Phase 2	Wazuh SIEM Deployed	Complete
Phase 2	Dashboard Accessible	Complete
Phase 3	Wazuh Agent on Kali	Complete
Phase 3	Agent Active on Dashboard	Complete
Phase 4	SSH Brute Force Attack Simulated	Complete
Phase 4	Attack Detected on Dashboard	Complete
Phase 5	MITRE ATT&CK Mapping Viewed	Complete
Phase 5	Wazuh Security Report Exported	Complete
Phase 5	Incident Report Written	Complete
Phase 5	Project Published on GitHub	Complete